Notation and Preliminary.

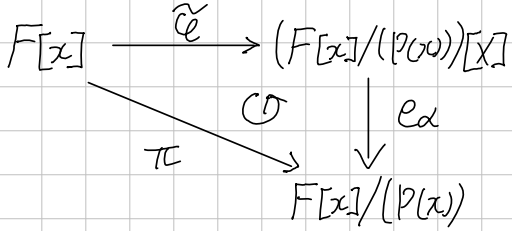
Let F_1, F_2 be fields, and let $\varphi: F_1 \to F_2$ be a field embbedding.

$\widetilde{\varphi}: F_1[x] \hookrightarrow F_2[x]: P(x) \mapsto P^{\varphi}(x)$ where $P(x)=\sum_{k=0}^{n} a_k \cdot x^k$ and $P^{\varphi}(x)=\sum_{k=0}^{n} \varphi(a_k) \cdot x^k$.

And, for $\alpha \in F$, an evaluation as

$$e_\alpha: F[x] \to F: P(x) \mapsto P(\alpha).$$

Now, let $P(x) \in F[x]$ be an irreducible. Then, for $\alpha = x + (P(x))$,



Lemma. Let R_1, R_2 be Commutative Rings with unity.
Let $\varphi: R_1 \to R_2$ be a Ring-Homomorphism. Then,

$$\varphi_x : R_1[x] \to R_2[x] : \sum_{i=0}^{n} a_i \cdot x^i \mapsto \sum_{i=0}^{n} \varphi(a_i) \cdot x^i$$

is a Ring-Homomorphism. Moreover,
1) φ_x is injective if and only if φ is injective
2) φ_x is surjective if and only if φ is surjective.

Proof. Let $\sum_{i=0}^{n} a_i \cdot x^i, \sum_{i=0}^{m} b_i \cdot x^i \in R_1[x]$ be given. Then,

$$\begin{aligned}
\varphi_x\left(\sum_{i=0}^{n} a_i \cdot x^i + \sum_{i=0}^{m} b_i \cdot x^i\right) &= \varphi_x\left(\sum_{i=0}^{\max\{n,m\}} (a_i + b_i) \cdot x^i\right) \\
&= \sum_{i=0}^{\max\{n,m\}} \varphi(a_i + b_i) \cdot x^i \\
&= \sum_{i=0}^{\max\{n,m\}} \left[\varphi(a_i) + \varphi(b_i)\right] \cdot x^i \\
&= \sum_{i=0}^{n} \varphi(a_i) \cdot x^i + \sum_{i=0}^{m} \varphi(b_i) \cdot x^i \\
&= \varphi_x\left(\sum_{i=0}^{n} a_i \cdot x^i\right) + \varphi_x\left(\sum_{i=0}^{m} b_i \cdot x^i\right)
\end{aligned}$$

$$\begin{aligned}
\varphi_x\left(\left[\sum_{i=0}^{n} a_i \cdot x^i\right] \cdot \left[\sum_{i=0}^{m} b_i \cdot x^i\right]\right) &= \varphi_x\left(\sum_{i=0}^{n+m} \left(\sum_{k=0}^{i} a_{i-k} \cdot b_k\right) \cdot x^i\right) \\
&= \sum_{i=0}^{n+m} \varphi\left(\sum_{k=0}^{i} a_{i-k} \cdot b_k\right) \cdot x^i \\
&= \sum_{i=0}^{n+m} \sum_{k=0}^{i} \left[\varphi(a_{i-k}) \cdot \varphi(b_k)\right] \cdot x^i \\
&= \left(\sum_{i=0}^{n} \varphi(a_i) \cdot x^i\right) \cdot \left(\sum_{i=0}^{m} \varphi(b_i) \cdot x^i\right) \\
&= \varphi_x\left(\sum_{i=0}^{n} a_i x^i\right) \cdot \varphi_x\left(\sum_{i=0}^{m} b_i \cdot x^i\right)
\end{aligned}$$

Thm. Let F be a field and let $p(x) \in F[x]$ be an irreducible polynomial. Then, there exists a field K containing F which has a root of $p(x)$.

Proof. Since $F[x]$ is P.I.D, an irreducible element $p(x)$ generates a maximal ideal. Therefore, the quotient $\overline{K} := F[x]/(p(x))$ is a field.
Define a map $\varphi : F \to \overline{K} : a \mapsto a + (p(x))$.

Then, φ is a field-Homomorphism. Since nonzero field Homomorphism must be injective, so φ is an embedding.
And, to show the last assertion, denote that $p(x) = \sum_{i=0}^{n} a_i x^i$.
By definition, $p^{\varphi}(x) = \sum_{i=0}^{n} \varphi(a_i) \cdot x^i$ and therefore

$$p^{\varphi}(\overline{\alpha}) = \sum_{i=0}^{n} \varphi(a_i) \cdot \overline{\alpha}^i = \sum_{i=0}^{n} \left(a_i + (p(x))\right) \cdot \left(x + (p(x))\right)^i$$
$$= \left(\sum_{i=0}^{n} a_i \cdot x^i\right) + (p(x)) = p(x) + (p(x)) = 0 + (p(x)).$$

Now, define $K := F \cup (\overline{K} \setminus \varphi[F])$, and let

$$h : \overline{K} \to K : u \mapsto \begin{cases} \varphi|_F^{-1}(u) & \text{if } u \in \varphi[F] \\ u & \text{if } u \notin \varphi[F]. \end{cases}$$

Then, h is bijective. Moreover, define two operations on K as

$$a + b := h(h^{-1}(a) + h^{-1}(b)) \quad \text{and} \quad a \cdot b := h(h^{-1}(a) \cdot h^{-1}(b))$$

Then, h is field-Homomorphism, so Proved. ☒

Note: we will write $h(f(x) + (p(x))) \in K$ as $f(\alpha) \bmod p(x)$.

Thm. Let $P(x) \in F[x]$ be an irreducible polynomial of degree n over a field F. Let K be the isomorphic copy of $F[x]/(P(x))$ and let $\alpha = x \bmod P(x) \in K$. Then, $\{1, \alpha, \alpha^2, \cdots, \alpha^{n-1}\}$ is a basis for K over F.